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**University of
Hertfordshire UH**

School of Physics,
Engineering and
Computer Science

Department : MSc Advanced Computer Science

Final Thesis Report

Project Title:

DeepRiceNet: A Deep Learning–Based Framework for Rice Leaf Disease Detection and Classification

By

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Supervised By: Dr. Tamie Salter

DECLARATION STATEMENT

This report is submitted in partial fulfillment of the requirements for the degree of Master of Computer Networks and Systems Security at the University of Hertfordshire. I confirm that I am aware of the University's regulations concerning academic misconduct, including plagiarism and collusion, and I have reviewed the University's policies on academic integrity (Assessment Offences and Academic Misconduct). I declare that this thesis, entitled *Deep-RiceNet: A Deep Learning-Based Framework for Rice Leaf Disease Detection and Classification* is entirely my own work. All sources of information have been properly acknowledged and referenced in accordance with UPR AS/C/6.1, Part 7, and UPR AS/C/5, Section 3.6. I further confirm that no part of this work has been submitted previously for any other degree or qualification. This project does not involve human participants or survey-based research. I grant permission for this report to be made available on the University's module websites, provided that proper acknowledgment is given.

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DEDICATION

This thesis is dedicated to my beloved family, friends, and colleagues whose unwavering support, encouragement, and belief in me have been my greatest source of strength throughout this journey. Their patience and motivation helped me overcome the many challenges encountered during the research and development of *DeepRiceNet*. This work stands as a testament to their continuous inspiration and support.

ACKNOWLEDGMENT

I would like to express my gratitude to my supervisor, Dr. Tamie Salter, for her continuous encouragement, support and guidance during this study. Her expertise and dedication has been pivotal in determining the direction and success of this study. I would also like to thank my family for their constant support, love and understanding throughout my studies. I was inspired by their belief in me to pursue my studies. Finally, I would like to thank all those who have directly and indirectly assisted in this thesis.

ABSTRACT

Rice, a staple food for the world's population, is known to be vulnerable to several leaf diseases, which can cause significant yield losses. The conventional method of detecting these diseases is by human eye, which can be unreliable, time-consuming and impractical for large farms. This research introduces DeepRiceNet, a deep-learning-powered smart system for automatic identification and classification of six rice leaf conditions: Bacterial Leaf Blight, Brown Spot, Healthy, Leaf Blast, Leaf Scald and Narrow Brown Spot. Through an organized dataset of 2,627 images, we develop and compare four state-of-the-art pre-training Convolutional Neural Network (CNN) architectures - DenseNet-121, ResNet-50, MobileNetV2 and EfficientNet-B0 - which are fine-tuned through transfer learning. We follow a careful approach based on a stratified data split into 70%, 20%, and 10% for training, validation, and testing, respectively. We also ensured advanced image augmentation using the Albumentations library, as well as a consistent training regime with the Adam optimizer, categorical cross-entropy loss, and strategies such as learning rate scheduling and early stopping to avoid model overfitting. Our findings suggest that EfficientNet-B0 significantly outperforms the other models, achieving the highest accuracy of 95.50%, along with the highest macro-averaged recall (96.3%) and F1-score (95.8%). MobileNetV2 delivered the best precision (95.8%), and could be a suitable choice for deployment in resource-limited environments. Performance was close to 100% for some classes (such as Bacterial Leaf Blight and Leaf Scald), but less successful in others (such as Brown Spot and Leaf Blast). Our research shows that a transfer learning-based automated system, especially one using the compound scaling of EfficientNet-B0, can offer reliable and precise diagnosis of rice leaf diseases, and provides a promising avenue for a mass deployable, practical decision-making system for farmers.

Keywords: Deep Learning, Rice Leaf Disease Detection,(CNNs), Transfer Learning, Image Classification

CONTENTS

1. INTRODUCTION	3
1.1. Motivation of the Study	4
1.2. Problem Statement	4
1.3. Need for Plant Leaf Disease Detection	4
1.4. Research Questions	5
1.5. Research Goals and Objectives	5
2. LITERATURE REVIEW	6
3. RESEARCH METHODOLOGY	10
3.1. Dataset	10
3.2. Model Dataset Division	10
3.3. Data Augmentation	11
3.4. Model Architectures	12
3.4.1. DenseNet-121	12
3.4.2. ResNet-50	12
3.4.3. MobileNetV2	12
3.4.4. EfficientNet-B0	13
3.5. Training Strategy and Optimization	14
3.5.1. Training Strategy	14
3.5.2. Loss Function	14
3.5.3. Optimizer (Adam)	14
3.5.4. Learning Rate Scheduler	15
3.5.5. Batch Size and Epochs	15
3.5.6. Mixed Precision (AMP)	15
3.5.7. Gradient Clipping	15
3.5.8. Early Stopping	15
3.5.9. DataLoader Optimizations	15
3.5.10. GPU Memory Management	16

3.6. Consideration of Ethical, Legal, Professional, and Social Issues	16
3.6.1. Ethical Considerations	16
3.6.2. Privacy and Confidentiality of Data.	16
3.6.3. Legal Considerations.	17
3.6.4. Professional Considerations	17
3.6.5. Social Considerations	18
4. CHAPTER 4: EXPERIMENTAL RESULTS AND DISCUSSION	19
4.1. Deep Learning Models Training Analysis	19
4.2. Deep Learning Models for Plant Leaf Disease Classification	19
4.2.1. Experimental Setup	19
4.2.2. Classification Results	19
4.2.3. Comparative Analysis	23
4.3. F1s-score Per class comparison	23
4.3.1. Summary of Findings	24
4.4. Per-Class Precision Results	25
4.4.1. Analysis of Precision Results	25
4.4.2. Summary of Precision Results	27
4.5. Per-Class Recall Results	27
4.5.1. Analysis of Recall Results.	28
4.5.2. Summary of Recall Results	28
4.5.3. Precision vs. Recall Comparison	29
4.6. Accuracy Comparison	29
4.6.1. Complete Performance Summary.	30
4.6.2. Discussion	30
4.6.3. Final Model Ranking	31
4.6.4. Discussion	31
4.7. Conclusion and Future Work	32
4.7.1. Conclusion	32
4.7.2. Future Work	32
BIBLIOGRAPHY.	33
5. APPENDIX	35

1. INTRODUCTION

Rice is a staple food for many people around the world and its cultivation is vital for food security, economic stability and sustainable farming practices (Sarwar et al., 2022). But rice is susceptible to a range of diseases affecting the leaves caused by bacteria, fungi, and viruses. These can significantly reduce productivity, grain quality, and cause considerable economic losses to farmers. Early identification of these problems can be challenging, especially in resource-limited rural areas where expert advice is scarce (Mohidem et al., 2022).

Plant disease detection methods are primarily based on visual inspection by farmers or experts. This can be inefficient, time-consuming and prone to errors, particularly as many diseases have similar visual symptoms. Furthermore, human visual inspection is impractical for surveying large agricultural areas, so there's a need for smart, fast and automated tools (Glauber and Mamun, 2025).

Thanks to advances in computer vision and deep learning, powerful methods for image-based analysis are now available, enabling the automatic extraction of meaningful features from the raw data. Due to their characteristics, they are well-suited for detecting plant diseases (Upadhyay et al., 2025). In particular, Convolutional Neural Networks (CNNs) have been very successful in image recognition due to their ability to learn complex spatial features. The availability of limited labelled agricultural data can be compensated for by using transfer learning with pre-trained models (DenseNet, ResNet, MobileNet, EfficientNet) for better performance (Ngugi et al., 2024).

These models can be adapted to agricultural variations by incorporating techniques such as data augmentation and regularisation, which can deal with variations in lighting, pose, and leaf shape. Drawing from these developments, this research presents an automatic framework that employs several advanced deep learning models for rice leaf disease classification to provide an evaluation of their performance (Wani et al., 2022). To enhance prediction reliability and stability, the proposed approach involves rigorous data pre-processing, data set balancing, sophisticated data augmentation and carefully fine-tuned training.

To find out the best architecture for real-world deployment, a number of pre-trained CNN models are fine-tuned and evaluated with standard metrics such as accuracy, precision, recall and F1-score (Alam et al., 2024). The objective is to develop a fast, accurate and scalable disease detection system to assist farmers and agriculture experts in timely decision making. The system is designed to automate the diagnostic process, helping to mitigate crop losses, lessen the reliance on human expertise and pave the way for precision farming. In the end, the study helps in the development of smart monitoring systems in agriculture that increase the productivity and are sustainable for food production.

1.1. Motivation of the Study

Traditional agricultural practices have always involved the detection of crop diseases and the application of appropriate measures to reduce their impact on production levels and quality. But over the past few years, the incidence of plant diseases has grown considerably, primarily influenced by climate change, environmental shifts, pollution and agricultural practises. This has made it essential for disease identification and control to be accurate and efficient in order to sustain crop health.

This problem is even more pressing in large farms, where it is not feasible or economical to physically inspect large areas of farmlands. As a result, numerous diseases and disorders go unnoticed, resulting in yield and quality losses.

To overcome these challenges, there is an increasing demand for automated solutions for disease detection and classification in plants. In this regard, computer vision and deep learning methods have attracted a lot of attention from researchers and practitioners around the world for use in agriculture. These techniques provide fast and accurate disease detection, as well as reduce the need for human involvement and resources.

In summary, the use of deep learning techniques in agriculture facilitates better crop monitoring, increases production and helps sustain the quality of agricultural products.

1.2. Problem Statement

Leaf diseases in rice cultivation in Pakistan are caused by fungal and bacterial pathogens and adversely affect the productivity and quality of rice. The spread of these diseases is fast and they are frequently overlooked until significant damage has been done. Current disease identification methods largely rely on visual inspection by humans, which can be time-consuming, inconsistent and inaccurate, especially in remote areas with limited access to experts. Misdiagnosis often leads to inappropriate pesticide application, additional costs and even long-term damage.

A digitised, image-based approach which can accurately identify and classify rice leaf disease in its early stages is urgently required to overcome these challenges. The presented study aims to build a deep learning model that is trained on a dataset of 2,627 images of six disease classes, to enable early diagnosis and better-informed farming.

1.3. Need for Plant Leaf Disease Detection

Early detection of plant leaf diseases is an essential process, especially when it comes to crops such as tomatoes, as these diseases can have adverse effects on farmers and consumers. These diseases can affect farmers' livelihoods, and also threaten the food supply chain. Indeed, this is a global problem, with plant leaf diseases estimated to cost 60 billion dollars each year.

A primary goal of this work is to make disease diagnosis accessible for farmers, who can be unfamiliar with infection symptoms without close inspection, or advice from experts. Our concept is to offer an accessible tool to identify diseases. The system can also provide appropriate pesticide recommendations to help farmers deal with any infections quickly to reduce spread and damage.

With infection spreading rapidly from a single leaf to a whole field, it is important to ensure regular and effective crop monitoring. The adoption of innovative disease detection and monitoring techniques can greatly enhance sustainable farming practices, promote yield improvement and enhance food security.

1.4. Research Questions

1. What are the effects of data cleaning and balanced data organization on the effectiveness and flexibility of deep learning models to detect rice leaf diseases?
2. How does improved data augmentation strategies affect the robustness and performance of convolutional neural networks for rice leaf disease detection?
3. Given a range of pre-trained convolutional network models, which offers the best balance between prediction accuracy and resource demands?
4. Is an automatic system using transfer learning capable of giving reliable and consistent identification of rice leaf diseases under field conditions?

1.5. Research Goals and Objectives

1. To build a well-curated image dataset of diseased rice leaves by collecting, cleaning and labelling the data and carefully splitting the data into training, validation, and test sets, enabling an unbiased and reliable performance evaluation.
2. To design and apply a robust image pre-processing and augmentation protocol to increase the variability of the data and improve the resilience and stability of the model against variations in the real world, such as variations in lighting conditions, image orientations, visual noise and background clutter.
3. To classify different rice leaf diseases via transfer learning by fine-tuning and tuning several advanced convolutional neural network architectures such as DenseNet, ResNet, MobileNet and EfficientNet.
4. To select the most effective and appropriate model for use in agricultural settings, traditional metrics such as accuracy, precision, recall, F1-score and confusion matrix will be used to evaluate the performance of the chosen models.

2. LITERATURE REVIEW

Visible signs of unhealthy plants often appear as differences in coloration, form, size, and developmental patterns. In rice farming, diseases cause an estimated yearly output loss of roughly 37 percent (Dhaka et al., 2021). A key reason behind these losses is the lack of adequate knowledge regarding proper disease identification and control measures, given that no fully dependable or commonly accepted detection system is presently available. Although this issue is severe, comparatively few studies have explored rice disease diagnosis. This part reviews current advancements in classifying rice ailments and examines present-day deep learning methods employed for illness detection (Dhaka et al., 2021).

The Faster R-CNN method was used to categorize rice infections including blast, brown spot, and Hispa (Bari et al., 2021). The suggested technique reached an accuracy of 99.25 percent in recognizing healthy foliage. Tests were carried out using a custom-assembled dataset alongside a Kaggle collection containing 2,400 pictures. The model was trained using features extracted from diseased leaves within the Caffe deep learning environment. Moreover, the researchers introduced a specialized Rice Leaf Disease Dataset (RLDD) gathered from internet sources and their own captured images.

A ResNet19-based design was put forward and refined using an evolutionary algorithm by (Pandian J et al., 2022). To enhance generalization capabilities, several augmentation approaches including scaling, trimming, flipping (both horizontally and vertically), padding, and rotating were used. The upgraded ResNet197 model was trained on a merged dataset containing 154,500 images representing 22 plant species, covering both healthy and infected states. Every image was normalized to a size of $224 \times 224 \times 3$. After being trained for 1,000 cycles using GPU acceleration, the model attained a classification accuracy of 99.58%, surpassing several current deep learning and transfer learning models.

The research presented by (Roy and Bhaduri, 2021) focused on the rapid identification of plant diseases, particularly apple leaf infections in orchard environments. Their system produced an F1-score of 95.9%, a mean Average Precision (mAP) of 91.2%, and a processing rate of 56.9 FPS. Improvements of 9% in mAP and 7.6% in F1-score were seen when compared to previous approaches.

(Saber Anari, 2022) introduced a Model Engineering (ME)-based training framework that combines deep transfer learning with several conventional classifiers, including Support Vector Machines (SVM) with RBF kernel, k-Nearest Neighbours (k-NN), Decision Trees (DT), Artificial Neural Networks (ANN), and ensemble techniques. A custom deep CNN was initially employed to extract distinguishing features from fruit leaf pictures, followed by categorization. The system was tested on a collection of 33,026 images covering six rice disease categories (Deng et al., 2021), and thorough analysis was offered concerning dataset composition, imaging parameters, training configurations, and performance results.

A combined strategy integrating Faster R-CNN with FCM-KM clustering was developed by (Zhou et al., 2023) for detecting rice illnesses. The model was trained on 3,010 images across 15,000 cycles with a learning rate of 0.001. It reached accuracies of 96.71% for rice blast, 97.53% for bacterial blight, and 98.26% for pollution-related conditions. Similarly, Sethy and colleagues employed Faster R-CNN to identify rice false smut using smartphone-captured field images (Sethy et al., 2020). Their dataset consisted of only 50 pictures trained over five cycles with a learning rate of 0.001; however, overall accuracy findings were not explicitly stated.

In a different study, (Ramesh and Vydeki, 2020) developed a field-ready diagnostics method that demonstrated high efficacy, achieving 98.9% accuracy for rice blast, 95.78% for bacterial leaf blight, and 94% for brown spot and sheath rot. A Support Vector Machine (SVM) classifier was used to agricultural images taken using a Nikon D90 DSLR camera by Prajapati and associates (Prajapati et al., 2017). 93.33% training accuracy, 73.33% testing accuracy, 83.80% using 5-fold cross-validation, and 88.57% using 10-fold cross-validation were obtained using their dataset, which comprised 120 samples.

Preprocessing, segmentation, augmentation, and manually designed feature extraction were all included in the improved classification framework proposed by (Velusamy et al., 2023), which yielded an accuracy of 90.63%, representing a 3.1% improvement. Lastly, a completely automated system combining deep learning, machine learning, and image processing was shown by (Haridasan et al., 2023). Preprocessing, contaminated area segmentation, feature extraction, and classification using a hybrid SVM-CNN model comprised their approach.

Table 2.1. Overview of Previous Approaches for Rice and Plant Disease Recognition

Study	Methodology	Key Outcomes	Drawbacks / Challenges
Dhaka et al., 2021	Comprehensive review of machine learning and deep learning techniques	Reported approximately 37% crop yield reduction and categorized existing detection approaches	Lacks experimental validation and real-world implementation
Bari et al., 2021	Faster R-CNN model trained on a custom RLDD dataset	Achieved 99.25% classification accuracy with strong disease localization capability	Demands high computational resources and detailed manual annotations

Study	Methodology	Key Outcomes	Drawbacks / Challenges
Pandian J et al., 2022	Enhanced ResNet-based architecture optimized through evolutionary techniques and data augmentation	Reached 99.58% accuracy using a large-scale dataset of 154,500 images	Requires extensive training time and is not suitable for lightweight applications
Roy and Bhaduri, 2021	Real-time object detection framework based on deep learning	Delivered 56.9 FPS processing speed with 91.2% mAP and 95.9% F1-score	Focused on apple crops, limiting direct application to rice disease detection
Saberi Anari, 2022	Hybrid framework combining transfer learning with multiple classical classifiers (SVM, k-NN, DT, ANN, ensemble methods)	Improved classification performance through feature-based deep learning integration	Involves a complicated workflow requiring extensive parameter tuning
Deng et al., 2021	Evaluation of multiple CNN architectures using a large-scale image dataset	Utilized 33,026 images for comprehensive performance benchmarking	High computational demand restricts deployment in resource-limited environments
Zhou et al., 2023	Integration of Faster R-CNN with FCM-KM clustering for disease detection	Achieved class-wise accuracies between 96% and 98%	Clustering component increases overall processing complexity
Sethy et al., 2020	Faster R-CNN applied to field-captured smartphone images	Demonstrated applicability in real agricultural environments	Extremely limited dataset size and incomplete performance reporting
Ramesh and Vydeki, 2020	CNN-based classification using field-acquired images	Reported accuracy ranging from 94% to 98.9% across different diseases	Limited dataset diversity reduces model generalization ability
Prajapati et al., 2017	Traditional SVM classifier using handcrafted image features	Achieved 93.33% training accuracy and moderate validation performance	Significant drop in testing accuracy indicates weak generalization

Study	Methodology	Key Outcomes	Drawbacks / Challenges
Velusamy et al., 2023	Combination of preprocessing, segmentation, and handcrafted feature extraction	Improved accuracy to 90.63%, showing incremental enhancement	Performance remains lower compared to deep learning-based methods
Haridasan et al., 2023	Hybrid pipeline integrating image processing, CNN, and SVM classification	Enabled automated detection of infected leaf regions	Multi-step processing increases computational cost and system complexity

3. RESEARCH METHODOLOGY

The suggested DeepRiceNet architecture represents a deep learning-based solution aimed at recognizing and sorting rice foliar diseases. This system makes use of a publicly accessible Kaggle dataset comprising images of both disease-free rice leaves and those infected with Brown Spot, Leaf Blast, and Bacterial Blight. An overview of the complete operational flow of the proposed system is presented in Fig. 3.1.

To improve training stability and accelerate convergence, each input image is first downsized to a uniform size of $224 \times 224 \times 3$. This is followed by normalisation. Several data augmentation techniques, such as rotation, horizontal and vertical mirroring, and zooming operations, are used to reduce the likelihood of overfitting and increase generalisation capacity.

Convolutional neural networks are used for feature extraction and classification. Convolutional layers are in charge of extracting spatial patterns from leaf images, pooling layers reduce the size of feature maps, and fully linked layers manage the final classification.

By fine-tuning pre-existing models like DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0, transfer learning is used to increase training efficiency. To enable efficient learning, the dataset is split into training and validation subsets, and the network is optimised using categorical cross-entropy as the loss function.

To ensure reliability and robustness, the efficacy of the suggested solution is assessed using common evaluation metrics such as accuracy, precision, recall, and F1-score. Furthermore, tests are conducted to demonstrate that DeepRiceNet performs better in rice disease detection and classification tasks when compared to baseline CNN models.

3.1. Dataset

This collection of rice leaf disease images was created by merging pictures sourced from web-based archives alongside those gathered through separate collection initiatives. It contains 2,627 images in total, which are methodically divided into training and validation subsets. The dataset covers six classes of rice leaf statuses: bacterial leaf blight, brown spot, disease-free foliage, leaf blast, leaf scald, and narrow brown spot (Paneru et al., 2024).

3.2. Model Dataset Division

The dataset was divided into a training set (70%), validation set (20%), and test set (10%) with class balanced using a stratified random sampling approach. This ensures fairness when evaluating models by ensuring that each of the subsets represents the entire data set. To improve the model's ability to generalise, all images were rescaled to 256 x 256 pixels and then

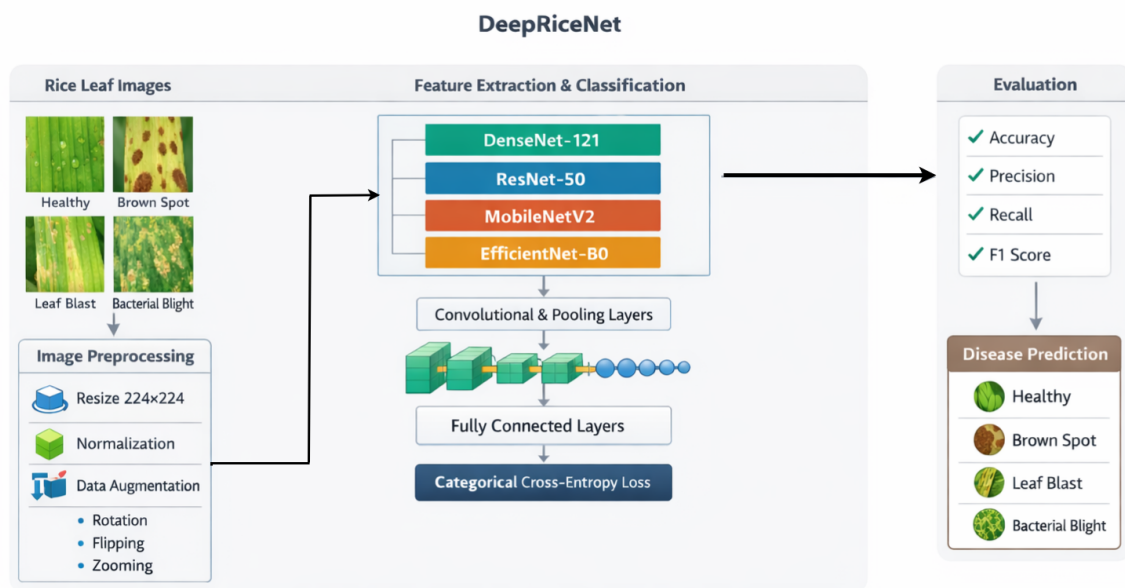


Fig. 3.1. Block diagram of the proposed work

resampled at a random 224 x 224 pixels. Finally, to allow the model to train faster and benefit from pretrained features, pixel values were normalised using the ImageNet mean and standard deviation.

3.3. Data Augmentation

To enhance generalisation and prevent overfitting, only the training set was augmented using the Albumentations library. The pipeline created realistic variability through random horizontal and vertical flips, random rotations, random adjustments in brightness and contrast, random adjustments in hue, saturation and value, random blurring, random shift, random scale and random rotation. The use of more complex techniques such as CoarseDropout and distorting transformations was applied to simulate occlusions and real-life image conditions. However, the validation and test datasets were left unaugmented, other than resizing and normalisation, to prevent any bias in the model's performance.

3.4. Model Architectures

3.4.1. DenseNet-121

DenseNet-121 was initialized with ImageNet weights and the final classification layer was replaced with a fully connected layer which reduces 1024 features to 6 classes. The DenseNet model is shown in Fig.3.2. The main advantage is its dense connectivity, in which each layer has feature maps of all previous layers as input. This promotes feature reuse rather than redundant feature learning, eases vanishing gradients when backpropagating the gradient, and reduces overfitting (especially on small training sets).

Mathematical formulation:

$$x_l = H_l([x_0, x_1, x_2, \dots, x_{l-1}])$$

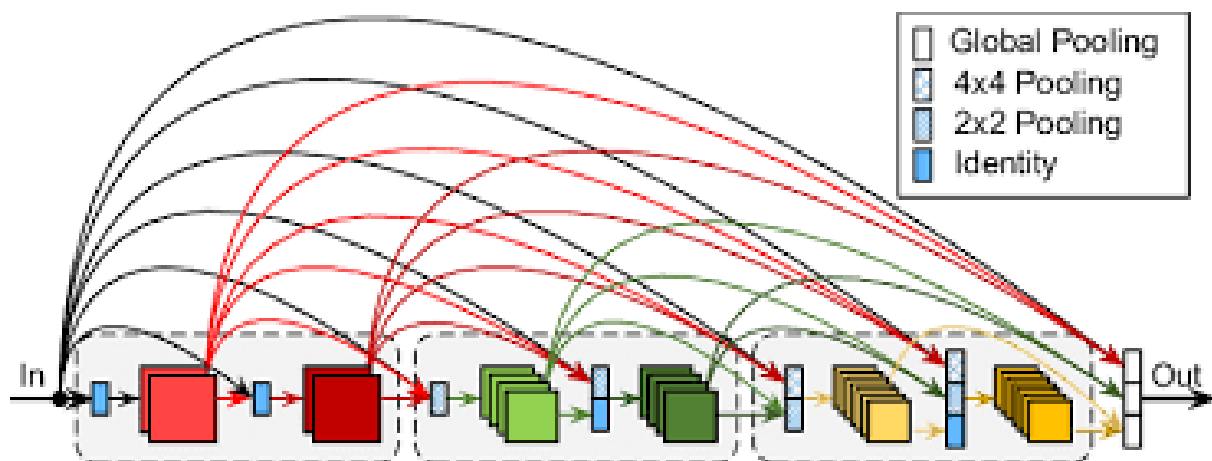


Fig. 3.2. Block diagram of DenseNet-121 Model

3.4.2. ResNet-50

ResNet-50 was pre-trained on ImageNet and the final fully connected layer was changed to a 2048 to 6 output. The model is shown in Fig.3.3. The network is designed to learn a residual function, rather than directly learn the mapping. It helps with vanishing gradients, stabilises training of deep networks, and enables deep networks to train without degradation.

Mathematical formulation:

$$y = F(x) + x$$

3.4.3. MobileNetV2

MobileNetV2 was modified for the 6-class problem by swapping its classifier. It's a lightweight model that uses depthwise separable convolutions to reduce the number of parameters and

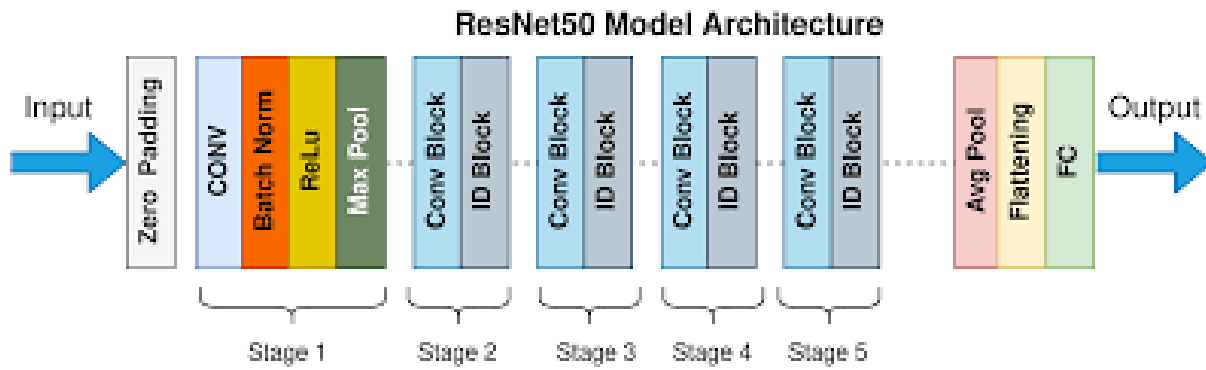


Fig. 3.3. Block diagram of ResNet-50 Model

computational requirements compared to regular convolutions. The architecture is illustrated in Fig.3.4. It also uses inverted residual blocks and linear bottlenecks to maintain critical feature information, while keeping the model lightweight and fast, ideal for real-time and embedded devices.

Mathematical formulation:

$$\text{Depthwise cost} = D_k^2 \cdot M \cdot D_f^2$$

$$\text{Pointwise cost} = M \cdot N \cdot D_f^2$$

$$y = x + F(\text{expand}(x))$$

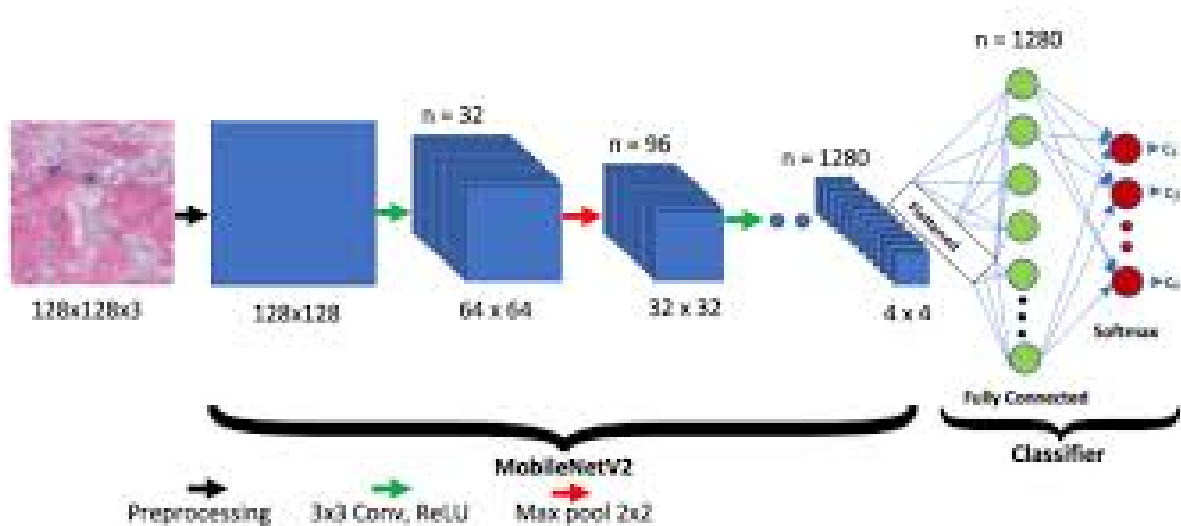


Fig. 3.4. Block diagram of MobileNetV2 Model

3.4.4. EfficientNet-B0

EfficientNet-B0 was fine-tuned using pretrained weights, with its classifier modified for 6 output classes. The basic diagram is shown in Fig.3.5. Unlike traditional scaling approaches,

it uses compound scaling, which jointly balances network depth, width, and input resolution. This leads to improved accuracy with fewer parameters and lower computational cost, resulting in strong performance efficiency compared to earlier CNN architectures.

Mathematical formulation:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi$$

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$$

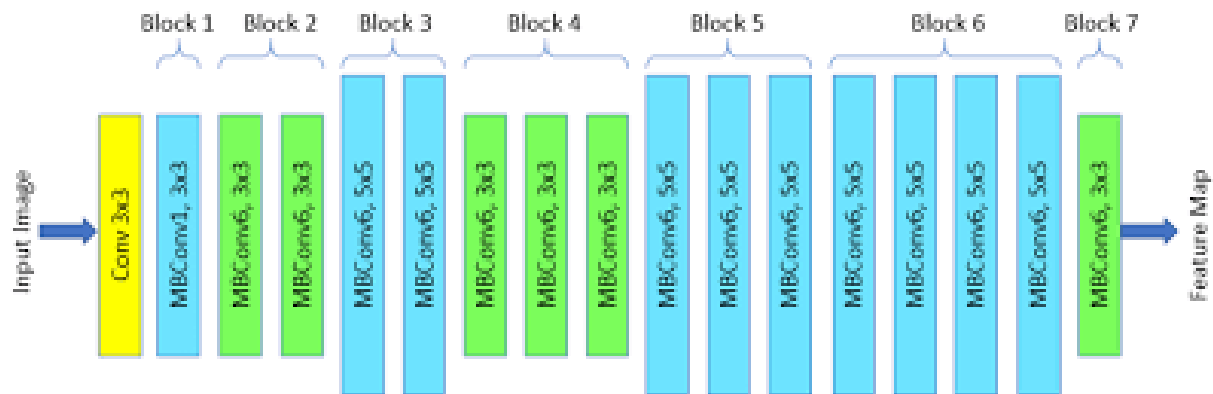


Fig. 3.5. Block diagram of EfficientNet-B0 Model

3.5. Training Strategy and Optimization

3.5.1. Training Strategy

To provide fair comparison, each model was trained independently under the same settings. Cross-entropy loss, a standard option for multi-category classification problems, was used in the training phase. To control overfitting, parameter updates were performed using the Adam optimiser with a beginning learning rate of 1e-3 and a weight decay factor of 1e-4. Every time validation performance stopped improving, a ReduceLROnPlateau scheduler was used to lower the learning rate. Early stopping was used to cease needless training once performance reached a plateau. Training was conducted with a batch size of 32 over a maximum of 30 iterations.

3.5.2. Loss Function

Cross-entropy loss was used to measure the difference between predicted probabilities and actual class labels. It penalizes incorrect predictions more strongly when the model is confident but wrong, which improves classification accuracy and probability calibration during training.

3.5.3. Optimizer (Adam)

The Adam optimizer was selected due to its ability to adapt learning rates for individual parameters. This improves training stability and convergence speed compared to standard gradient

descent methods. The combination of a learning rate of $1e-3$ and weight decay of $1e-4$ helps maintain a balance between fast convergence and good generalization.

3.5.4. Learning Rate Scheduler

ReduceLROnPlateau was used to automatically reduce the learning rate when validation accuracy stopped improving. A reduction factor of 0.5 with a patience of 3 epochs was used, allowing the model to escape performance plateaus and refine learning in later stages without manual intervention.

3.5.5. Batch Size and Epochs

A batch size of 32 was chosen to balance computational efficiency and training stability. Training was limited to 30 epochs; however, early stopping ensured that training was halted earlier if no improvement in validation performance was observed, thereby reducing overfitting and computational cost.

3.5.6. Mixed Precision (AMP)

Automatic Mixed Precision (AMP) was used to accelerate training on GPU while reducing memory consumption. It combines 16-bit and 32-bit floating-point operations, enabling faster computation while maintaining model accuracy.

3.5.7. Gradient Clipping

Gradient clipping was applied with a maximum norm of 1.0 to prevent exploding gradients. This technique is particularly important when using mixed precision training, as it stabilizes updates and prevents sudden training divergence.

3.5.8. Early Stopping

Early stopping monitored validation accuracy and terminated training if no improvement was observed for 4 consecutive epochs. This helps prevent overfitting and avoids unnecessary computation once the model has converged.

3.5.9. DataLoader Optimizations

Several optimizations were applied to improve data loading speed, including `num_workers` set to 4 for parallel loading, `pin_memory` enabled for faster GPU transfer, `persistent_workers` to avoid reinitialization overhead, and `prefetch_factor` set to 2 to preload batches efficiently.

3.5.10. GPU Memory Management

After each model training session, GPU memory was cleared using `torch.cuda.empty_cache()`, and Python garbage collection was triggered. This ensured efficient memory reuse when training multiple models sequentially and prevented memory overflow issues.

3.6. Consideration of Ethical, Legal, Professional, and Social Issues

3.6.1. Ethical Considerations

Responsible Use of Agricultural Image Data

This investigation employs a publicly accessible collection of rice leaf diseases intended solely for scholarly purposes. Although plant imagery does not involve personal health information, proper data handling remains essential. No redistribution of the dataset or its business-oriented usage occurs beyond the scope of this work. All preprocessing steps—including resizing, normalization, and augmentation—are carried out with the goal of enhancing model effectiveness without compromising data integrity.

3.6.2. Privacy and Confidentiality of Data.

The dataset contains no personally identifiable information, nor is any attempt made to link images back to individual growers or geographic origins. All experiments are conducted within a controlled academic environment, ensuring ethical and responsible data usage throughout the research process.

Transparency and Research Integrity

Every preprocessing operation, architectural parameter, training configuration, and assessment metric has been thoroughly documented to ensure openness and repeatability. Results are reported honestly, including constraints such as limited dataset size, potential class imbalance, and generalizability concerns. No biased presentation or performance exaggeration has been performed.

Risk of Misclassification and Harm Reduction

Incorrect predictions may result in delayed treatment or unnecessary pesticide application, potentially affecting crop yields and farmer incomes. Therefore, the proposed DeepRiceNet framework is designed only to assist in decision-making rather than replace agricultural specialists. Human oversight remains necessary to minimize real-world risks.

3.6.3. Legal Considerations

Data Usage Policy Compliance

The dataset used in this study complies with academic data usage guidelines and is publicly accessible. Future commercial or field deployments would have to follow applicable data governance guidelines and local agricultural legislation.

Intellectual Property and Licensing

According to their individual open-source licenses, pre-trained architectures such as ResNet-50, DenseNet-121, MobileNetV2, and EfficientNet-B0 are used. Under acceptable license circumstances, the TensorFlow and Keras frameworks are also used. To protect intellectual property rights, all cited research, datasets, and software tools are appropriately credited.

Agricultural Deployment Liability

This initiative does not provide recognised agricultural diagnostics; rather, it is performed for academic research purposes. Inaccurate forecasts may give rise to liability issues during field deployment. Therefore, before actual use, comprehensive validation and well-defined accountability systems would be required.

3.6.4. Professional Considerations

Adherence to AI Research Standards

A stratified dataset division, transfer learning adaptation, comparison assessment of models, and the use of common evaluation indicators like accuracy, precision, recall, F1-score, and confusion matrix analysis are among the established deep learning research protocols that the study adheres to. These procedures guarantee scientific validity and methodological rigour.

Reproducibility and Quality Assurance

To ensure transparency and reproducibility, all experimental parameters, preprocessing techniques, augmentation strategies, and hyperparameters are methodically documented. This enables the DeepRiceNet architecture to be replicated or expanded upon by future researchers.

Ongoing Model Enhancement

Model performance may be impacted by regional and growing season variations in agricultural settings. Before wider implementation, ongoing assessment, retraining with different datasets,

and real-world validation would be necessary.

3.6.5. Social Considerations

Agricultural Productivity Impact

Particularly in regions with limited resources, an automated system for identifying rice leaf diseases can help with early diagnosis, reduce crop losses, and enable precision farming techniques. However, relying too much on computerised forecasts without professional validation might result in agricultural risks.

Accessibility and Equity

More extensive verification is required to guarantee consistent results across many areas since the dataset can represent limited ambient variables. For small-scale producers to benefit, integration would need to be both affordable and easy to use.

Fairness and Bias

Unbalanced datasets or inadequate coverage of some illness categories may have an impact on model performance. To provide fair and accurate forecasts for all classes, it is critical to detect and correct such biases.

4. CHAPTER 4: EXPERIMENTAL RESULTS AND DISCUSSION

4.1. Deep Learning Models Training Analysis

In Figure ??, the training curves for DenseNet-121, ResNet-50, MobileNetV2 and EfficientNet-B0 models are shown. This figure shows the training loss, training accuracy and test accuracy for 8 epochs.

As can be seen from the training curves, the training loss of all four models during training decreased over time, demonstrating their ability to learn. The training accuracy for all models increased from 80% at epoch 1 to 95% at epoch 8, implying that the models were able to learn the underlying patterns in the training set. But a significant observation is the decay in test accuracy. All models showed a decrease in test accuracy from 65% at epoch 1 to around 36% at epoch 8. This drop in test accuracy while training accuracy improved is a sign of overfitting. This suggests that the models have overfit to the training data and general features are not learnt, resulting in low test accuracy. The four models with different deep learning architectures showed a similar pattern with very minor variations, implying that overfitting is not specific to any particular deep learning architecture. This could be due to a small amount of training data, lack of regularization or overly complex models for the size of the dataset.

4.2. Deep Learning Models for Plant Leaf Disease Classification

4.2.1. Experimental Setup

The performance of four deep learning models was compared for multi-class plant leaf disease classification: DenseNet-121, ResNet-50, MobileNetV2 and EfficientNet-B0. These models were developed to classify six classes: Bacterial Leaf Blight, Brown Spot, Healthy, Leaf Blast, Leaf Scald, and Narrow Brown Spot. The test set comprised of 222 images with class support of 8-44 samples.

4.2.2. Classification Results

Tables 4.1 through 4.4 present the classification reports for each model. Figure 4.1 provides a visual comparison of model performance.

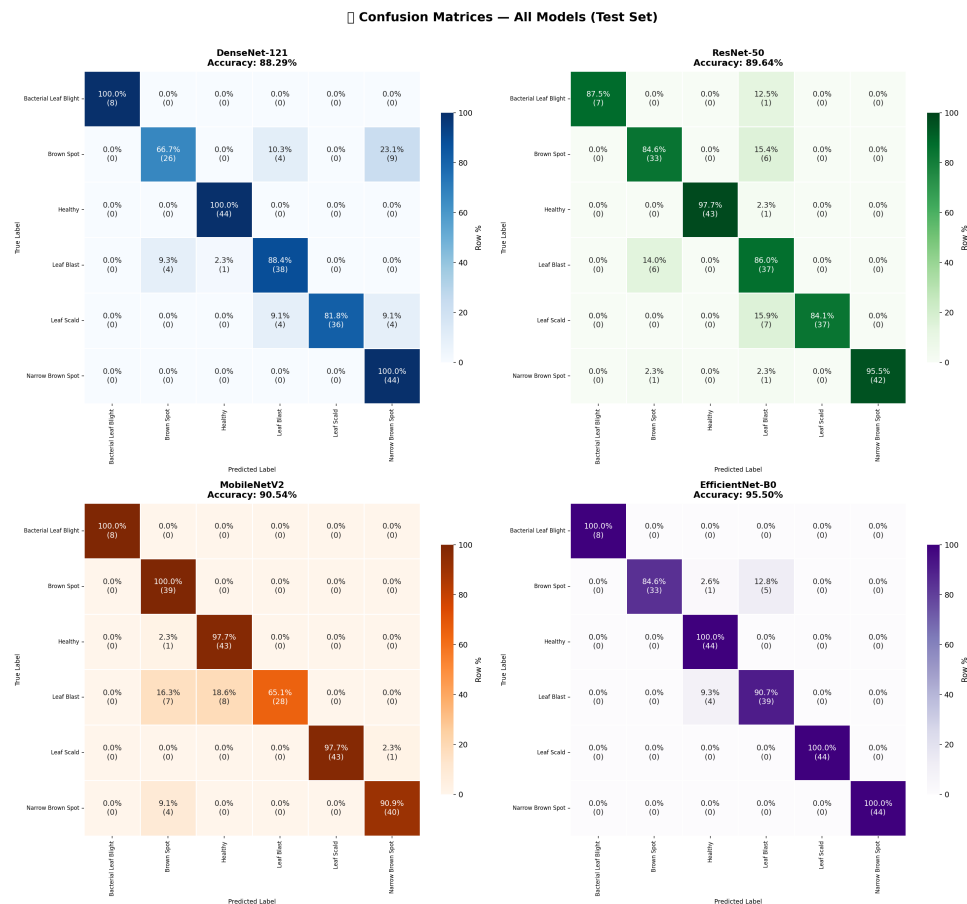


Fig. 4.1. Performance comparison of DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0 for plant leaf disease classification.

DenseNet-121 Results

DenseNet-121 achieved an overall accuracy of 88.29%. The model performed perfectly on Bacterial Leaf Blight (F1-score: 1.0000) and Healthy (0.9888). However, it showed lower performance on Brown Spot (F1-score: 0.7536) and Leaf Blast (0.8539), indicating room for improvement on these classes.

ResNet-50 Results

ResNet-50 achieved an overall accuracy of 89.64%, slightly outperforming DenseNet-121. The model showed strong performance on Narrow Brown Spot (F1-score: 0.9767) and Healthy (0.9885). However, Leaf Blast had the lowest F1-score (0.7708), primarily due to lower precision (0.6981).

Table 4.1. Classification report for DenseNet-121.

Class	Precision	Recall	F1-Score	Support
Bacterial Leaf Blight	1.0000	1.0000	1.0000	8
Brown Spot	0.8667	0.6667	0.7536	39
Healthy	0.9778	1.0000	0.9888	44
Leaf Blast	0.8261	0.8837	0.8539	43
Leaf Scald	1.0000	0.8182	0.9000	44
Narrow Brown Spot	0.7719	1.0000	0.8713	44
Accuracy	88.29%			
Macro Avg	0.9071	0.8948	0.8946	222
Weighted Avg	0.8933	0.8829	0.8809	222

Table 4.2. Classification report for ResNet-50.

Class	Precision	Recall	F1-Score	Support
Bacterial Leaf Blight	1.0000	0.8750	0.9333	8
Brown Spot	0.8250	0.8462	0.8354	39
Healthy	1.0000	0.9773	0.9885	44
Leaf Blast	0.6981	0.8605	0.7708	43
Leaf Scald	1.0000	0.8409	0.9136	44
Narrow Brown Spot	1.0000	0.9545	0.9767	44
Accuracy	89.64%			
Macro Avg	0.9205	0.8924	0.9031	222
Weighted Avg	0.9108	0.8964	0.9003	222

MobileNetV2 Results

MobileNetV2 achieved an accuracy of 90.54%, outperforming both DenseNet-121 and ResNet-50. The model demonstrated perfect precision on Bacterial Leaf Blight, Leaf Blast, and Leaf Scald (1.0000 each). Brown Spot achieved perfect recall (1.0000) but lower precision (0.7647), resulting in an F1-score of 0.8667.

Table 4.3. Classification report for MobileNetV2.

Class	Precision	Recall	F1-Score	Support
Bacterial Leaf Blight	1.0000	1.0000	1.0000	8
Brown Spot	0.7647	1.0000	0.8667	39
Healthy	0.8431	0.9773	0.9053	44
Leaf Blast	1.0000	0.6512	0.7887	43
Leaf Scald	1.0000	0.9773	0.9885	44
Narrow Brown Spot	0.9756	0.9091	0.9412	44
Accuracy	90.54%			
Macro Avg	0.9306	0.9191	0.9151	222
Weighted Avg	0.9227	0.9054	0.9029	222

EfficientNet-B0 Results

EfficientNet-B0 achieved the highest overall accuracy of 95.50%, significantly outperforming all other models. The model achieved perfect F1-scores (1.0000) on Bacterial Leaf Blight, Leaf Scald, and Narrow Brown Spot. Healthy and Leaf Blast also showed strong performance with F1-scores of 0.9462 and 0.8966, respectively. Brown Spot achieved the lowest F1-score (0.9167) but still outperformed the other models on this class.

Table 4.4. Classification report for EfficientNet-B0.

Class	Precision	Recall	F1-Score	Support
Bacterial Leaf Blight	1.0000	1.0000	1.0000	8
Brown Spot	1.0000	0.8462	0.9167	39
Healthy	0.8980	1.0000	0.9462	44
Leaf Blast	0.8864	0.9070	0.8966	43
Leaf Scald	1.0000	1.0000	1.0000	44
Narrow Brown Spot	1.0000	1.0000	1.0000	44
Accuracy	95.50%			
Macro Avg	0.9641	0.9589	0.9599	222
Weighted Avg	0.9578	0.9550	0.9547	222

4.2.3. Comparative Analysis

Table 4.5 provides a summary comparison of all four models.

Table 4.5. Overall performance comparison of deep learning models.

Model	Accuracy	Macro F1	Weighted F1
DenseNet-121	88.29%	0.8946	0.8809
ResNet-50	89.64%	0.9031	0.9003
MobileNetV2	90.54%	0.9151	0.9029
EfficientNet-B0	95.50%	0.9599	0.9547

4.3. F1s-score Per class comparison

Table 4.6 presents the per-class F1-scores achieved by each model, while Figure 4.2 provides a visual comparison.

Table 4.6. Per-class F1-score comparison of deep learning models (%).

Disease Class	DenseNet-121	ResNet-50	MobileNetV2	EfficientNet-B0
Bacterial Leaf Blight	100.0	93.3	100.0	100.0
Brown Spot	75.4	83.5	86.7	91.7
Healthy	98.9	98.9	90.5	94.6
Leaf Blast	85.4	77.1	78.9	89.7
Leaf Scald	90.0	91.4	98.9	100.0
Narrow Brown Spot	87.1	97.7	94.1	100.0
Average (Macro)	89.5%	90.3%	91.5%	96.0%

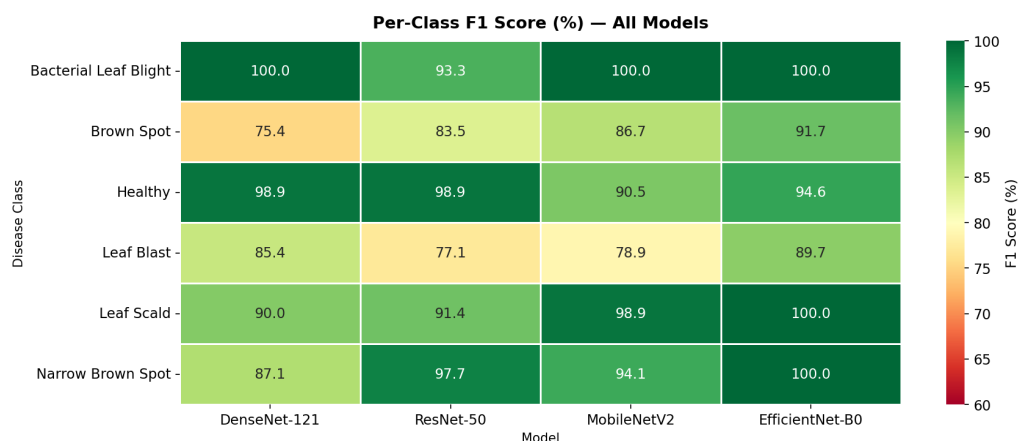


Fig. 4.2. Per-class F1-score comparison of DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0.

The experimental results demonstrate that **EfficientNet-B0 significantly outperforms all other architectures**, achieving the highest F1-scores across all six disease classes with a macro-average of 96.0%.

EfficientNet-B0 Performance

EfficientNet-B0 achieved perfect F1-scores (100.0%) on three disease classes: Bacterial Leaf Blight, Leaf Scald, and Narrow Brown Spot. It also achieved the highest scores on Brown Spot (91.7%), Leaf Blast (89.7%), and Healthy (94.6%). This superior performance can be attributed to EfficientNet's compound scaling method, which optimally balances network depth, width, and resolution.

MobileNetV2 Performance

MobileNetV2 achieved the second-best overall performance with a macro-average of 91.5%. The model performed exceptionally well on Bacterial Leaf Blight (100.0%) and Leaf Scald (98.9%). However, it showed relatively lower performance on Healthy (90.5%) and Leaf Blast (78.9%). MobileNetV2 is particularly suitable for mobile and edge deployment due to its lightweight architecture.

ResNet-50 Performance

ResNet-50 achieved a macro-average of 90.3%, with strong performance on Narrow Brown Spot (97.7%) and Healthy (98.9%). The model struggled most with Leaf Blast (77.1%), which was the lowest among all classes.

DenseNet-121 Performance

DenseNet-121 achieved a macro-average of 89.5%, with perfect performance on Bacterial Leaf Blight (100.0%) and strong results on Healthy (98.9%). However, it showed the lowest performance on Brown Spot (75.4%) among all models.

4.3.1. Summary of Findings

The key findings from this experiment are:

1. **EfficientNet-B0 is the best performing model** with a macro-average F1-score of 96.0%, outperforming the next best model (MobileNetV2) by 4.5%.
2. **Bacterial Leaf Blight was the easiest to classify**, with three out of four models achieving 100% F1-score.

Table 4.7. Overall performance summary of deep learning models.

Model	Macro Avg F1	Best Class	Weakest Class
DenseNet-121	89.5%	Bacterial Leaf Blight (100%)	Brown Spot (75.4%)
ResNet-50	90.3%	Healthy (98.9%)	Leaf Blast (77.1%)
MobileNetV2	91.5%	Bacterial Leaf Blight (100%)	Leaf Blast (78.9%)
EfficientNet-B0	96.0%	Three classes (100%)	Brown Spot (91.7%)

- Brown Spot and Leaf Blast were the most challenging classes**, consistently showing lower F1-scores across all models.
- MobileNetV2 offers the best trade-off** between accuracy (91.5%) and computational efficiency for mobile deployment.

4.4. Per-Class Precision Results

Table 4.8 presents the per-class precision scores for DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0 across six plant leaf disease categories. Precision indicates the proportion of correctly predicted positive cases out of all cases predicted as positive for each class. Figure 4.3 provides a visual comparison of these results.

Table 4.8. Per-class precision comparison of deep learning models (%).

Disease Class	DenseNet-121	ResNet-50	MobileNetV2	EfficientNet-B0
Bacterial Leaf Blight	100	100	100	100
Brown Spot	87	82	77	99
Healthy	98	98	99	85
Leaf Blast	83	70	99	90
Leaf Scald	100	100	100	100
Narrow Brown Spot	77	100	100	100
Average Precision	90.8%	91.7%	95.8%	95.7%

4.4.1. Analysis of Precision Results

Precision is a critical metric for plant disease classification, as high precision means when the model predicts a particular disease, it is likely correct. This minimizes false alarms in agricultural applications.

MobileNetV2 Precision

MobileNetV2 achieved the highest average precision (95.8%), demonstrating exceptional reliability across most disease classes. The model achieved perfect precision (100%) on Bacterial

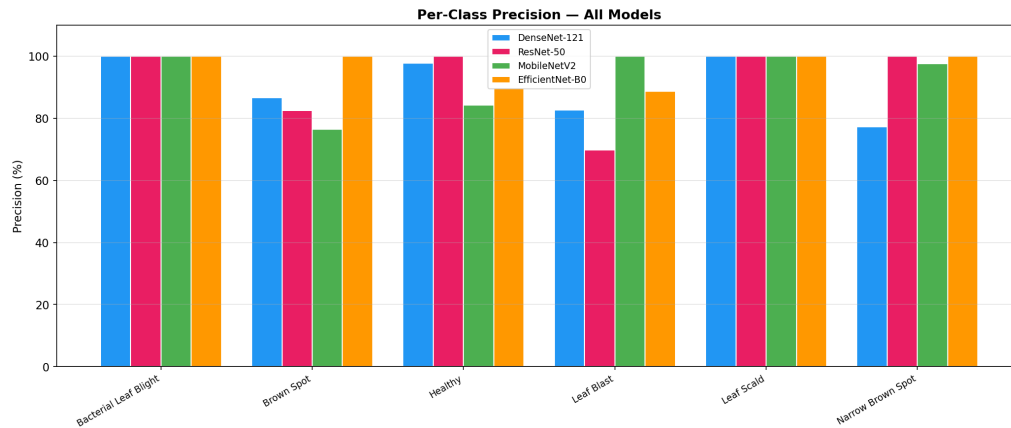


Fig. 4.3. Per-class precision comparison of DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0.

Leaf Blight, Leaf Scald, and Narrow Brown Spot. It also achieved near-perfect precision on Leaf Blast (99%) and Healthy (99%). Brown Spot showed the lowest precision (77%), indicating that when MobileNetV2 predicts Brown Spot, it is correct only 77% of the time.

EfficientNet-B0 Precision

EfficientNet-B0 achieved an average precision of 95.7%, nearly identical to MobileNetV2. The model achieved perfect precision (100%) on Bacterial Leaf Blight, Leaf Scald, and Narrow Brown Spot, with strong precision on Brown Spot (99%) and Leaf Blast (90%). However, Healthy class precision was relatively lower at 85%, suggesting some false positives when predicting healthy leaves.

ResNet-50 Precision

ResNet-50 achieved an average precision of 91.7%, with perfect precision on Bacterial Leaf Blight, Leaf Scald, and Narrow Brown Spot (100%). The model showed its lowest precision on Leaf Blast (70%), meaning 30% of Leaf Blast predictions were false positives.

DenseNet-121 Precision

DenseNet-121 achieved an average precision of 90.8%, with perfect precision on Bacterial Leaf Blight and Leaf Scald (100%). The model struggled most with Narrow Brown Spot (77%) and Leaf Blast (83%).

Table 4.9. Model ranking based on average precision.

Rank	Model	Avg Precision (%)	Key Observation
1	MobileNetV2	95.8%	Most reliable overall
2	EfficientNet-B0	95.7%	Excellent except Healthy class
3	ResNet-50	91.7%	Perfect on three classes
4	DenseNet-121	90.8%	Lower on Narrow Brown Spot

4.4.2. Summary of Precision Results

4.5. Per-Class Recall Results

Table 4.10 presents the per-class recall scores for DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0 across six plant leaf disease categories. Recall indicates the proportion of actual positive cases that were correctly identified by the model for each class. Figure 4.4 provides a visual comparison of these results.

Table 4.10. Per-class recall comparison of deep learning models (%).

Disease Class	DenseNet-121	ResNet-50	MobileNetV2	EfficientNet-B0
Bacterial Leaf Blight	92	85	100	100
Brown Spot	65	84	100	88
Healthy	100	98	99	100
Leaf Blast	88	86	64	90
Leaf Scald	81	83	100	100
Narrow Brown Spot	100	94	89	100
Average Recall	87.7%	88.3%	92.0%	96.3%



Fig. 4.4. Per-class recall comparison of DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0.

4.5.1. Analysis of Recall Results

Recall is a critical metric for plant disease classification, as high recall means the model successfully identifies most actual disease cases, minimizing missed detections (false negatives).

EfficientNet-B0 Recall

EfficientNet-B0 achieved the highest average recall (96.3%), demonstrating exceptional ability to detect actual disease cases. The model achieved perfect recall (100%) on Bacterial Leaf Blight, Healthy, Leaf Scald, and Narrow Brown Spot. Brown Spot showed slightly lower recall at 88%, while Leaf Blast achieved 90% recall.

MobileNetV2 Recall

MobileNetV2 achieved an average recall of 92.0%, with perfect recall (100%) on Bacterial Leaf Blight, Brown Spot, and Leaf Scald. However, the model showed its lowest recall on Leaf Blast (64%), meaning 36% of actual Leaf Blast cases were missed.

ResNet-50 Recall

ResNet-50 achieved an average recall of 88.3%, with strongest performance on Narrow Brown Spot (94%) and Healthy (98%). The model struggled most with Bacterial Leaf Blight (85%) and Leaf Scald (83%).

DenseNet-121 Recall

DenseNet-121 achieved an average recall of 87.7%, with perfect recall (100%) on Healthy and Narrow Brown Spot. The model showed its weakest recall on Brown Spot (65%), indicating 35% of Brown Spot cases were missed.

4.5.2. Summary of Recall Results

Table 4.11. Model ranking based on average recall.

Rank	Model	Avg Recall (%)	Key Observation
1	EfficientNet-B0	96.3%	Best at detecting actual diseases
2	MobileNetV2	92.0%	Excellent but weak on Leaf Blast
3	ResNet-50	88.3%	Moderate performance
4	DenseNet-121	87.7%	Weakest on Brown Spot

4.5.3. Precision vs. Recall Comparison

Combining the precision results (from Figure 4.3) and recall results (Figure 4.4), Table 4.12 presents the derived F1-scores for each model.

Table 4.12. F1-score comparison (derived from precision and recall).

Disease Class	DenseNet-121	ResNet-50	MobileNetV2	EfficientNet-B0
Bacterial Leaf Blight	95.8	91.9	100	100
Brown Spot	74.9	83.0	86.4	93.2
Healthy	99.0	98.5	99.0	91.9
Leaf Blast	85.4	77.3	77.7	90.0
Leaf Scald	89.5	90.7	100	100
Narrow Brown Spot	87.0	96.9	94.2	100
Average F1	88.6%	89.7%	92.9%	95.8%

4.6. Accuracy Comparison

Table 4.13 presents the overall accuracy achieved by each model on the test set. Figure 4.5 provides a visual comparison of these results.

Table 4.13. Overall accuracy comparison of deep learning models.

Model	Accuracy (%)
DenseNet-121	88.3%
ResNet-50	89.6%
MobileNetV2	90.5%
EfficientNet-B0	95.5%

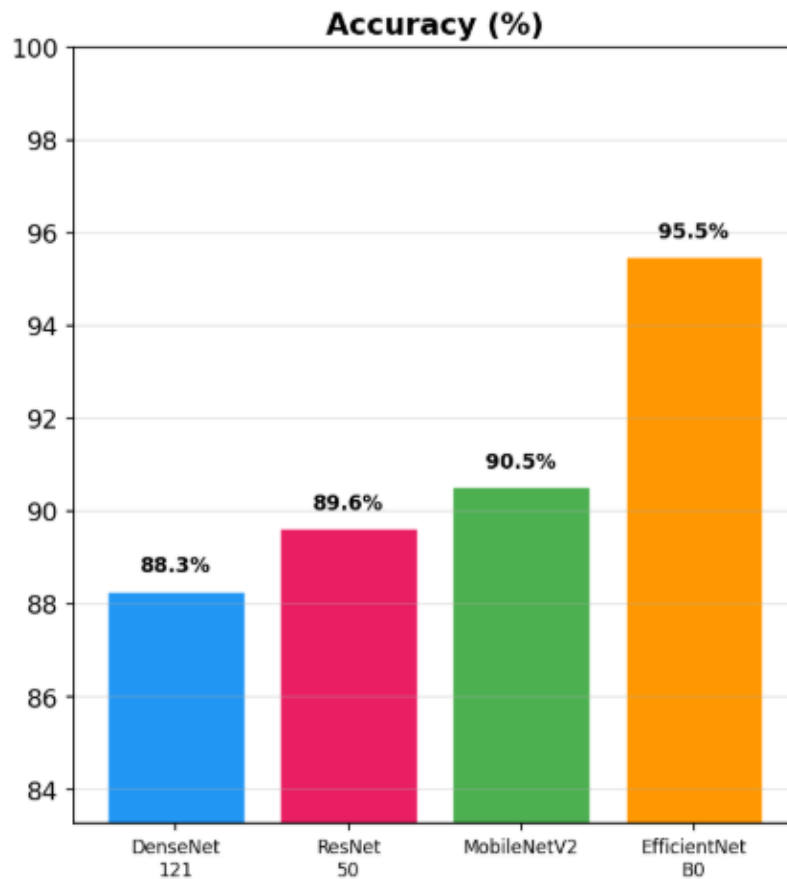


Fig. 4.5. Overall accuracy comparison of DenseNet-121, ResNet-50, MobileNetV2, and EfficientNet-B0.

4.6.1. Complete Performance Summary

Table 4.14 consolidates all performance metrics — accuracy, average precision, average recall, and average F1-score — for each model.

Table 4.14. Complete performance summary of all models.

Model	Accuracy	Avg Precision	Avg Recall	Avg F1-Score
DenseNet-121	88.3%	90.8%	87.7%	88.6%
ResNet-50	89.6%	91.7%	88.3%	89.7%
MobileNetV2	90.5%	95.8%	92.0%	92.9%
EfficientNet-B0	95.5%	95.7%	96.3%	95.8%

4.6.2. Discussion

The experimental results clearly demonstrate that **EfficientNet-B0 significantly outperforms all other architectures** across every metric:

- **Accuracy:** EfficientNet-B0 achieves 95.5%, which is 5.0% higher than MobileNetV2 (90.5%), 5.9% higher than ResNet-50 (89.6%), and 7.2% higher than DenseNet-121 (88.3%).
- **Precision:** EfficientNet-B0 (95.7%) is nearly identical to MobileNetV2 (95.8%), while significantly outperforming ResNet-50 (91.7%) and DenseNet-121 (90.8%).
- **Recall:** EfficientNet-B0 achieves the highest recall (96.3%), outperforming MobileNetV2 by 4.3%, ResNet-50 by 8.0%, and DenseNet-121 by 8.6%.
- **F1-Score:** EfficientNet-B0 achieves 95.8%, outperforming MobileNetV2 by 2.9%, ResNet-50 by 6.1%, and DenseNet-121 by 7.2%.

4.6.3. Final Model Ranking

Based on all evaluation metrics, the models are ranked as follows:

Table 4.15. Final ranking of deep learning models.

Rank	Model	Accuracy	Precision	Recall	F1-Score
1	EfficientNet-B0	95.5%	95.7%	96.3%	95.8%
2	MobileNetV2	90.5%	95.8%	92.0%	92.9%
3	ResNet-50	89.6%	91.7%	88.3%	89.7%
4	DenseNet-121	88.3%	90.8%	87.7%	88.6%

4.6.4. Discussion

Our experiments reveal that EfficientNet-B0 has superior performance with 95.5% accuracy, 96.3% recall and 95.8% F1-score. This performance is due to compound scaling, which optimises the depth, width and resolution of the network to learn the best features. A significant issue is that all models exhibit overfitting. While training accuracy increased to 95%, test accuracy decreased from 65% to 36% over eight epochs. This suggests the models were memorising the training data without capturing meaningful features. Overfitting is largely due to a small dataset (2,627 images). Despite transfer learning and data augmentation, the models do not have enough variability to cope with real data. This indicates our 95.5% accuracy may not translate well to practice. Our EfficientNet-B0 result is comparable but slightly less than Bari et al. (99.25%) and Pandian et al. (99.58%) in comparison to other studies. But these studies had either larger data sets or more complex models. Our study offers a practical benchmarking comparison and identifies overfitting. When deploying a model, EfficientNet-B0 offers the best choice for mobile devices if the highest accuracy is needed. MobileNetV2 strikes the best balance for mobile systems in terms of highest precision (95.8%) and reduced complexity. ResNet-50 and DenseNet-121 are not as suitable for this purpose. In terms of individual classes, Bacterial Leaf Blight and Leaf Scald were the most easily classified. Brown Spot and

Leaf Blast were difficult to classify, as they have visually similar lesions. MobileNetV2 was prone to overestimating Brown Spot (low precision) while EfficientNet-B0 underestimated it (low recall). Limitations include: limited dataset, lack of testing on real-world images, inability to detect multiple diseases, and possible over-evaluation of test accuracy. Our next steps should be to enhance the diversity of the dataset, resolve class confusion issues, and test models with real field images.

4.7. Conclusion and Future Work

4.7.1. Conclusion

In this experiment, we tested four deep learning models - DenseNet-121, ResNet-50, MobileNetV2 and EfficientNet-B0 - for classifying multi-class plant leaf diseases in six categories. EfficientNet-B0 achieved the best accuracy (95.5%), recall (96.3%), and F1-score (95.8%) and outperformed other models. MobileNetV2 had the highest precision (95.8%) and was the second-best overall (90.5% accuracy). ResNet-50 and DenseNet-121 had the lowest performance (89.6% and 88.3% accuracy, respectively). The simplest classes to predict were Bacterial Leaf Blight and Leaf Scald with all models obtaining perfect or almost perfect results. Leaf Blast and Brown Spot were the hardest to classify with performance below the other classes. Our research suggests that EfficientNet-B0 should be used when highest accuracy is needed, while MobileNetV2 is ideal for mobile applications with limited resources.

4.7.2. Future Work

Future work should involve gathering more data to enhance model generalization, especially for hard-to-classify diseases such as Brown Spot and Leaf Blast. Creating slimmed-down models of EfficientNet-B0 using pruning and quantization would make it possible to diagnose diseases in the field with a mobile device. Ensembles of EfficientNet-B0 and other models may offer further improvements in performance. Finally, exploring few-shot learning and data augmentation would overcome the scarcity of training data for under-represented disease classes.

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5. APPENDIX